

V01.09 – Mirror Physics III

Mirror Physics III

Thermodynamic Observerhood

Memory, Error Correction and the Energetic Cost of Persistence

Lloyd Christopher Smith

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Abstract

Mirror Theory treats observerhood as a derived property of bounded systems that maintain world-models, self-models, reliability variables and governed identity-continuity under viability constraints. Earlier work in Volume I formalised constraint regimes, minimal observer thresholds, recursive reliability, memory governance and the physical support conditions required for persistent observerhood. Mirror Physics I identified causal order, local substrate persistence, metastable memory, thermodynamic throughput and error-corrective repair as support conditions for observer-continuity. Mirror Physics II introduced constraint salience as a scale- and horizon-relative organisational quantity, while refusing to treat it as a new field or force. This paper develops the thermodynamic side of that sequence. It argues that an observer is not merely an information-processing structure, but a thermodynamically maintained constraint structure: it must spend physical resources to preserve memory, correct error, maintain boundaries, regulate update and sustain identity across time. The paper introduces a persistence-cost functional decomposed into memory, correction, boundary, update and audit costs. It states a thermodynamic observerhood condition: reliability tracking and self-model governance become observer-relevant only when their expected viability contribution exceeds their physical and computational maintenance cost. The analysis is deliberately conservative. It does not modify thermodynamics, relativity or quantum theory. It identifies the cost structure that any physically realised observer must satisfy.

Keywords: Mirror Theory; thermodynamic observerhood; persistence cost; memory; error correction; boundary maintenance; identity-continuity; reliability; viability; constraint salience; thermodynamics; information; maintenance; observerhood.

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1 Introduction

Mirror Theory begins with a methodological refusal: observerhood should not be assumed at the foundation of explanation. It should be derived, or at least constrained, from the kinds of bounded systems that can persist, model, correct and update themselves within a structured world. The earlier Mirror papers developed this claim in stages. The theory arc treated observerhood as a result of distinguishability, bounded modelling, viability, self-modelling and recursive reliability. The mathematical arc introduced constraint regimes, minimal observer thresholds and memory-governed identity-continuity. The computational laboratory sequence then tested these ideas in toy agents under perturbation, showing that reliability variables are useful only when they are action-guiding, cost-sensitive and coupled to viability.

The Physics Support Arc adds a conservative physical question. Mirror Physics I asked what physical reality must provide for observer-continuity to be realised. Mirror Physics II asked whether the organisation supporting such continuity can be assigned a measurable salience quantity. The present paper asks what such persistence costs physically.

If an observer is not a momentary label but a continuity-preserving process, then observerhood is not free. A persistent observer must preserve information against noise. It must correct error. It must maintain a boundary, at least functionally, between self-relevant and environment-relevant variables. It must update without indiscriminate self-overwrite. It must dissipate energy, consume free energy or otherwise maintain itself against thermal, computational and environmental degradation.

The guiding claim is therefore:

An observer is not merely an information-processing structure. It is a thermodynamically maintained constraint structure that must spend physical resources to preserve memory, correct error, regulate update and maintain identity across time.

This claim is not a modification of established physics. It does not assert a new force, field, conservation law, spacetime geometry or exception to thermodynamic principles. Rather, it identifies the physical cost conditions that any persistent observer must satisfy. In this sense, Mirror Physics III is deliberately modest: it is a bridge paper, not a replacement theory.

The central mathematical object is the persistence-cost functional

$$K_L^{\text{persist}} = K_L^{\text{memory}} + K_L^{\text{correction}} + K_L^{\text{boundary}} + K_L^{\text{update}} + K_L^{\text{audit}}. \quad (1)$$

This expression is not intended as a microscopic law. It is a bookkeeping structure for the physical and computational costs of maintaining observer-continuity. Its importance lies in its connection to the Mirror salience condition:

$$\Sigma_L(R_L) = \mathbb{E}[V_L \mid M_E, M_L, R_L, \pi_R] - \mathbb{E}[V_L \mid M_E, M_L, \pi_0] - K_L(R_L, \pi_R). \quad (2)$$

Reliability tracking is not valuable merely because it is recursive or elegant. It is valuable when it improves expected viability after its cost is counted. The same is true of memory governance, error correction and identity maintenance. The thermodynamic interpretation of Mirror Theory is therefore cost-sensitive from the start.

2 Position within the Physics Support Arc

This paper assumes the following Volume I sequence, as situated by the Volume I overview [1].

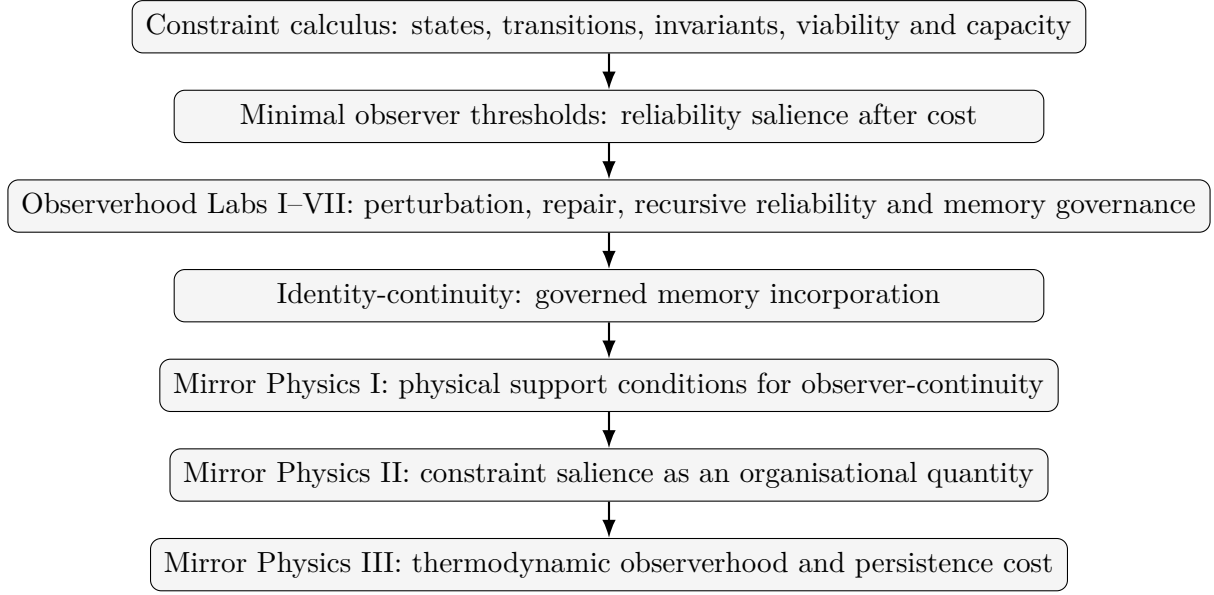


Figure 1: The position of Mirror Physics III within the Volume I sequence. The present paper adds the cost structure required for physically maintained observer-continuity.

Mirror Mathematics I defined a constraint regime as a structured tuple of states, transitions, invariants, viability and bounded capacity [2]. Mirror Mathematics II stated the minimal observer threshold: a system becomes observer-like only when self-model reliability is represented, action-guiding, cost-sensitive and positively coupled to viability under perturbation [3]. Mirror Mathematics III extended the analysis to identity-continuity: memory is not identity; identity is governed continuity of a self-model under perturbation [4].

The Observerhood Labs supplied computational scaffolding [7–13]. They did not establish consciousness, nor did they prove Mirror Theory in the physical world. They showed, in controlled toy environments, how self-model reliability, channel-decomposed reliability, cost-sensitive repair, recursive reliability and memory governance behave under perturbation. These studies suggested that recursive structure is beneficial only under specific conditions: when the perturbation is relevant, the diagnosis is actionable and the expected benefit exceeds the cost.

Mirror Physics I identified the physical support conditions required for observer-continuity: causal order, finite propagation, memory substrates, thermodynamic throughput and error correction [5]. Mirror Physics II introduced constraint salience as a candidate organisational quantity, while carefully refusing to identify it with a new physical field [6]. The present paper now asks what it costs for a physical system to remain an observer across time.

3 Observerhood as maintained constraint

3.1 Local physical systems

Let L denote a local system embedded in an environment E . The system may be biological, artificial, social, computational or hybrid. The analysis does not require that L be conscious. It requires only that L possess physically realised states and transitions.

Let $X_L(t)$ be the physical state of L at time t . Let $B_L(t)$ denote its boundary conditions, broadly understood: membranes in cells, skin and sensory boundaries in organisms, access controls in software agents, institutional boundaries in scientific communities and so on. Let $M_L(t)$ denote memory-bearing states, and let $I_L(t)$ denote the identity-relevant self-model state introduced in Mirror Mathematics III [4].

A local observer-candidate over horizon τ may be represented as

$$O_L^\tau = (X_L, B_L, M_L, I_L, R_L, \mu_L, V_L), \quad (3)$$

where R_L is a reliability variable, μ_L is a maintenance policy and V_L is a viability functional defined over the relevant horizon.

Definition 1 (Thermodynamic observer-candidate). *A thermodynamic observer-candidate is a local physical system L for which there exists a horizon $\tau > 0$ such that L maintains memory-bearing, self-modelling and reliability-bearing states under a maintenance policy μ_L , and such maintenance is relevant to the expected viability of L over τ .*

This definition deliberately avoids claims about subjective experience. It concerns physical support for persistent observerhood, not phenomenology.

3.2 Maintenance as active preservation

Persistence is not passive. Even if a system appears stable, its stability is supported by some combination of physical structure, environmental regularity, repair, redundancy, low temperature, chemical throughput, computational refresh or institutional practice.

In Mirror terms, maintenance is the set of operations by which a system prevents its identity-relevant organisation from being dissolved by noise, drift, corruption or uncontrolled update. We write

$$\mu_L : (X_L, M_L, I_L, R_L, E_t) \mapsto A_L, \quad (4)$$

where A_L is a maintenance action or policy response. Examples include cellular repair, immune response, sleep-mediated memory consolidation, database checksums, RAID redundancy, model auditing, sensory recalibration or simply refusing to commit an unreliable memory.

Remark 1. *A maintenance policy need not be explicit or conscious. A bacterium, a brain, a server cluster and a language-agent memory system can all implement maintenance without representing it linguistically.*

4 Memory is physical

The Mirror account of identity relies on memory governance, but memory is not abstract in its realised form. A memory must be encoded somewhere: in molecular states, synaptic weights, magnetic domains, charge distributions, transistor states, written symbols, social practices or other durable physical arrangements.

4.1 Memory substrates

Let M_L denote the memory substrate of L . A memory substrate is useful only if it has enough stability to preserve distinctions over the relevant horizon:

$$\mathbb{P}(d(M_L(t), M_L(t + \tau)) < \epsilon) \geq q, \quad (5)$$

where d is a task-relevant distance measure, ϵ is an allowable distortion tolerance and q is the required preservation probability.

This condition already has thermodynamic content. Without some stability mechanism, memory states are vulnerable to noise, decay or overwrite. Long-horizon observerhood requires either robust physical encoding or active refresh.

4.2 Landauer caveat

Landauer’s principle states that logically irreversible erasure of one bit of information has a lower-bound heat cost of $k_B T \ln 2$ in ideal conditions [17]. Mirror Theory does not require that every memory operation be logically irreversible, nor does it equate all cognitive cost with Landauer erasure. The principle is relevant more modestly: memory manipulation, reset, correction and governance are not physically free when implemented in real substrates.

The paper therefore uses Landauer-style reasoning only as a constraint on physical realisation, not as a complete theory of cognition. Real systems incur additional costs: sensing, actuation, refresh, redundancy, computation, cooling, metabolism and communication.

Definition 2 (Memory maintenance cost). *The memory maintenance cost $K_L^{\text{memory}}(t, t + \tau)$ is the expected physical and computational cost required to preserve memory-bearing distinctions of L within task-relevant tolerance over horizon τ .*

This cost may be metabolic in a biological system, electrical in a data centre, chemical in a cell or institutional in a scientific community. The common structure is preservation of distinctions against degradation.

5 Error correction and observer-continuity

5.1 Error as constraint drift

In Mirror Theory, error is not merely incorrect representation. It is drift between a model and the constraint structure that makes action viable. For observer-continuity, the most important errors are self-relevant: errors in location, capacity, memory, goal, boundary, reliability or identity-state.

Let $e_L(t)$ denote self-relevant error. The system need not eliminate all error. It must keep error below the threshold at which viability or identity-continuity collapses:

$$\mathbb{P}(e_L(s) \leq e_{\max} \text{ for all } s \in [t_0, t_0 + \tau]) \geq q. \quad (6)$$

Error correction is therefore not an optional cognitive flourish. It is a physical precondition for durable observerhood.

Definition 3 (Correction cost). *The correction cost $K_L^{\text{correction}}$ is the expected cost of detecting, localising and repairing errors in memory, self-model, boundary, sensor, map or reliability variables over a specified horizon.*

5.2 Error-correction support theorem

Theorem 1 (Error-correction support). *Let L be a local observer-candidate with nonzero probability of self-relevant perturbation over horizon τ . If observer-continuity requires self-relevant error to remain below threshold $e_{\max}$ with probability at least q , then either L implements an error-correction mechanism at the relevant scale, or observer-continuity fails whenever uncorrected perturbations accumulate beyond $e_{\max}$.*

Proof. Assume nonzero probability of perturbation and no correction mechanism. Then perturbations are either absent, ignored or passively absorbed. By assumption, over horizon τ there is nonzero probability that accumulated perturbation exceeds $e_{\max}$. On those trajectories, the required continuity condition fails. To keep failure probability below $1 - q$, the system must either reduce perturbation exposure, increase tolerance or implement correction. These are all error-management mechanisms at the relevant scale. Therefore observer-continuity requires error correction in the broad sense described. $\square$

The theorem is intentionally simple. Its value lies in preventing a common mistake: treating observerhood as a static property while ignoring the maintenance processes required for it to persist.

6 Boundary maintenance

An observer cannot maintain itself if nothing distinguishes self-relevant variables from everything else. This does not require an absolute metaphysical boundary. It requires a functional boundary: some distinction between variables that belong to the maintained organisation and variables treated as external, contextual or environmental.

Definition 4 (Functional boundary). *A functional boundary B_L is a rule, structure or process that distinguishes self-relevant variables of L from environmental variables for purposes of maintenance, update, repair and action.*

In a cell, the membrane is a literal boundary. In an organism, the boundary includes body, nervous system and regulatory systems. In an artificial agent, it may include authenticated memory stores, model weights, state variables, permissions and tool access. In a scientific community, it may include peer-review norms, archives, institutional roles and accepted methods.

Boundary maintenance has cost. Membranes require metabolic upkeep. Cybersecurity requires monitoring. Cognitive boundaries require attention and inhibition. Institutions require enforcement. A system that cannot maintain its boundary is vulnerable to uncontrolled overwrite, capture or dissolution.

Definition 5 (Boundary maintenance cost). *The boundary maintenance cost K_L^{boundary} is the expected cost of preserving the functional distinction between self-relevant and environment-relevant variables over a specified horizon.*

Proposition 1 (Boundary requirement). *If no functional boundary exists for L , then there is no well-defined local target for observer-continuity. Without a local target, memory governance and self-model reliability cannot be assigned to L rather than to an arbitrary larger system.*

This proposition is not a claim that boundaries are sharp. Boundaries can be fuzzy, nested and scale-relative. The claim is only that some boundary condition is necessary for local observerhood to be defined.

7 Update without identity collapse

Mirror Mathematics III introduced memory governance because memory alone does not produce identity [4]. A system that incorporates every candidate memory becomes contaminated; a system that incorporates none becomes rigid. Thermodynamic observerhood adds that both governance failures have costs.

Let G_L be a memory governance function:

$$G_L(m_t, R_m, I_L, V_L) \in \{\text{commit, quarantine, audit, reject}\}. \quad (7)$$

Governance consumes resources. Auditing takes time. Verification consumes energy. Refusal may prevent useful adaptation. Commitment may reduce uncertainty but risks contamination. Thus update is not merely logical; it is thermodynamic and computational.

Definition 6 (Update cost). *The update cost K_L^{update} is the expected cost of incorporating reliable self-relevant information while preserving identity-continuity.*

Definition 7 (Audit cost). *The audit cost K_L^{audit} is the expected cost of evaluating the reliability of candidate self-relevant updates before incorporation into persistent memory or identity structure.*

The Observerhood Labs repeatedly show the same structure: reliability and audit improve viability only in regimes where the avoided damage exceeds the audit and repair cost [7–13]. This is the computational analogue of the thermodynamic principle developed here.

8 The persistence-cost functional

8.1 Cost decomposition

We now collect the maintenance costs:

$$K_L^{\text{persist}}(t, t + \tau) = K_L^{\text{memory}} + K_L^{\text{correction}} + K_L^{\text{boundary}} + K_L^{\text{update}} + K_L^{\text{audit}}. \quad (8)$$

A more detailed continuous-time version is

$$K_L^{\text{persist}}(t, t + \tau) = \mathbb{E} \left[\int_t^{t+\tau} \left(\dot{K}^{\text{memory}} + \dot{K}^{\text{correction}} + \dot{K}^{\text{boundary}} + \dot{K}^{\text{update}} + \dot{K}^{\text{audit}} \right) ds \right]. \quad (9)$$

This expression is scale-relative. In a cell, the terms refer to biochemical costs. In an artificial agent, they may refer to compute, storage, energy, latency and verification. In a social epistemic system, they may refer to institutional labour and communication costs. Mirror Theory does not collapse these into one physical unit prematurely; it demands that any concrete application specify the substrate and measurement regime.

8.2 Thermodynamic salience condition

Let μ_L be the active maintenance policy and μ_0 a baseline policy without the relevant maintenance mechanism. Define the thermodynamic salience of maintenance as

$$\Theta_L^\tau(\mu_L) = \mathbb{E}[V_L(t + \tau) \mid \mu_L] - \mathbb{E}[V_L(t + \tau) \mid \mu_0] - K_L^{\text{persist}}(t, t + \tau). \quad (10)$$

Definition 8 (Thermodynamic observerhood condition). *A maintenance mechanism is thermodynamically observer-salient over horizon τ when*

$$\Theta_L^\tau(\mu_L) > 0. \quad (11)$$

This is the physical analogue of the Mirror salience condition. It says that maintaining memory, reliability or identity is not automatically justified. It becomes observer-relevant only where the expected viability advantage exceeds maintenance cost.

Theorem 2 (Thermodynamic salience theorem). *Let L be an observer-candidate with viability functional V_L and maintenance policy μ_L . If the expected viability gain from maintenance is less than or equal to the persistence cost, then the maintenance mechanism is not thermodynamically observer-salient over that horizon.*

Proof. By definition,

$$\Theta_L^\tau(\mu_L) = \mathbb{E}[V_L \mid \mu_L] - \mathbb{E}[V_L \mid \mu_0] - K_L^{\text{persist}}.$$

If the viability gain is less than or equal to K_L^{persist} , then $\Theta_L^\tau(\mu_L) \leq 0$. Therefore the mechanism fails the thermodynamic observerhood condition. $\square$

This theorem is almost tautological, but the consequence is not trivial: Mirror Theory predicts conditional usefulness, not universal usefulness. Recursive reliability, memory governance and identity maintenance can become maladaptive if they are too expensive, too slow, too rigid or insufficiently targeted.

9 No zero-cost persistence

A system can appear persistent because its substrate is stable over the observed horizon. But physical persistence is never metaphysically costless. Either the substrate has been physically arranged to resist decay, or the system actively refreshes itself, or the horizon is short relative to the decay timescale.

Theorem 3 (No zero-cost persistence under noise). *Let L maintain identity-relevant distinctions over horizon τ in an environment with nonzero perturbation probability. If those distinctions are required to remain within tolerance ϵ with probability at least q , then either: (i) L relies on a physically stable substrate whose preparation and boundary conditions carry cost; (ii) L performs active maintenance with nonzero expected cost; or (iii) the continuity condition fails for sufficiently long horizons.*

Proof. If the substrate is not perfectly stable and perturbations occur with nonzero probability, then identity-relevant distinctions drift over time. To prevent drift beyond tolerance, the system must either be encoded in a substrate with physical stability relative to the horizon or be actively corrected. Stable encoding itself is not costless in physical preparation, boundary condition or environmental control. If neither stable encoding nor active maintenance is present, then over sufficiently long horizons accumulated perturbation exceeds tolerance on some non-negligible set of trajectories, violating the continuity condition. $\square$

The theorem expresses a simple but important physical point: persistence is a maintained achievement, not an ontological default.

10 The maintenance loop

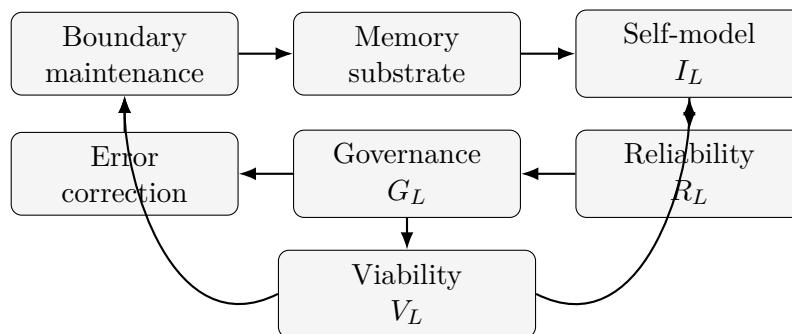


Figure 2: A schematic maintenance loop. Persistent observerhood requires boundary maintenance, memory preservation, self-model continuity, reliability estimation, governance, error correction and viability feedback. The diagram is conceptual and not a proposed physical mechanism.

The loop illustrates why observerhood cannot be reduced to storage, prediction or self-reference alone. A memory system without governance is contaminable. A self-model without reliability is brittle. Reliability without action is inert. Error correction without viability relevance is mere repair. Boundary maintenance without update becomes rigid. Observerhood emerges, in the Mirror sense, when these functions are coupled into a persistence-preserving loop.

11 Trade-off between contamination and rigidity

Memory governance introduces a trade-off. A system too open to update is vulnerable to contamination. A system too closed to update becomes rigid and loses adaptive capacity.

Let $\rho \in [0, 1]$ denote update openness, where $\rho = 0$ corresponds to complete refusal of self-relevant updates and $\rho = 1$ corresponds to indiscriminate incorporation. Let $C_{\text{contam}}(\rho)$ be contamination cost and $C_{\text{rigid}}(\rho)$ be rigidity cost. A generic governance cost can be represented as

$$C_{\text{gov}}(\rho) = C_{\text{contam}}(\rho) + C_{\text{rigid}}(\rho) + C_{\text{audit}}(\rho). \quad (12)$$

Mirror Theory predicts that persistent observerhood lives in an intermediate regime: enough openness to learn, enough governance to resist corruption and enough audit to distinguish the two without exhausting viability.

Proposition 2 (Governance optimum). *If contamination cost increases with openness and rigidity cost decreases with openness, then under mild continuity conditions there exists an intermediate update policy that outperforms both indiscriminate update and total refusal whenever audit cost does not dominate the avoided loss.*

This proposition abstracts the pattern observed in the memory-governance laboratory results [13]. The Mirror agent is not superior because it never updates. It is superior because it filters self-relevant updates through reliability-governed commitment.

12 Relation to constraint salience

Mirror Physics II introduced a candidate scale- and horizon-relative constraint salience quantity [6]:

$$\Phi_L^{\Gamma, \tau}. \quad (13)$$

This quantity was not proposed as a new physical field. It was introduced as a bridge concept: a possible way of measuring organisational structure relevant to persistence, reliability and observerhood.

Thermodynamic observerhood now adds that salience must be cost-sensitive. A highly complex structure that cannot maintain itself under perturbation has low observer-relevant salience. A simple structure that robustly preserves viability-relevant distinctions may have high salience at its scale.

A thermodynamically corrected salience functional may therefore take the schematic form

$$\Phi_{L, \text{therm}}^{\Gamma, \tau} = \alpha \mathcal{I}_L^{\Gamma, \tau} + \beta \mathcal{C}_L^{\Gamma, \tau} + \gamma \mathcal{E}_L^{\Gamma, \tau} + \delta \mathcal{R}_L^{\Gamma, \tau} - \lambda K_L^{\text{persist}}. \quad (14)$$

Here $\mathcal{I}_L^{\Gamma, \tau}$ may represent information-bearing structure, $\mathcal{C}_L^{\Gamma, \tau}$ causal coupling or partial closure, $\mathcal{E}_L^{\Gamma, \tau}$ error-correction capacity, $\mathcal{R}_L^{\Gamma, \tau}$ recursive reliability and K_L^{persist} the cost of maintenance. The coefficients are not universal constants. They belong to a chosen modelling scale, measurement regime and horizon.

The negative cost term is essential. Without it, the framework would overvalue arbitrarily elaborate systems. Mirror Theory instead asks whether organisation contributes to persistence after maintenance cost is paid.

13 Examples across scales

System	Persistence mechanisms	Mirror interpretation
Cell	Membrane, metabolism, DNA repair, homeostasis	Minimal biological observer-candidate if self-relevant regulation contributes to viability; not necessarily conscious.
Animal nervous system	Sensory integration, memory consolidation, motor control, immune and metabolic regulation	Stronger observer-candidate because self-state, action and error correction are coupled to survival.
Human person	Autobiographical memory, bodily regulation, social identity, narrative repair, metacognition	Rich identity-continuity with multiple memory-governance layers.
Artificial language agent	Persistent memory store, commit gate, tool verification, self-state tracking, audit policy	Observer-like only if self-model reliability is action-guiding, cost-sensitive and viability-relevant.
Scientific community	Archives, peer review, replication, correction, retraction, methodological norms	Distributed epistemic continuity; not automatically a conscious observer, but a constraint-maintaining system.

Table 1: Examples of thermodynamic or resource-costly persistence across scales. The table does not equate these systems phenomenologically. It identifies analogous maintenance structures.

The scale-relative nature of the analysis is important. A cell, a human and a scientific community do not share the same phenomenology. But each can be studied as a system that preserves information-bearing organisation through maintenance, correction and boundary regulation.

14 Relation to existing frameworks

14.1 Thermodynamics and life

The connection between life and thermodynamic organisation is old. Schrödinger famously described living systems as feeding on negative entropy, or more carefully, maintaining order through exchanges with the environment [19]. Prigogine’s work on dissipative structures emphasised how ordered patterns can arise and persist far from equilibrium [18]. Mirror Theory does not replace these ideas. It asks when such maintained organisation becomes observer-like through self-modelling, reliability tracking and identity governance.

14.2 Information and computation

Shannon information theory supplies tools for quantifying uncertainty and information transmission [20]. Landauer and Bennett connect information processing to physical cost [15,17]. Mirror Theory uses these resources but does not reduce observerhood to information quantity alone. Observerhood requires information organised into viability-relevant, self-relevant and reliability-governed persistence.

14.3 Cybernetics and control

Cybernetics emphasises feedback, control and regulation [14,21]. Mirror Theory inherits this tradition but adds an observer threshold: not every regulator is an observer. A thermostat regulates, but it does not maintain a self-model whose reliability is action-guiding and identity-preserving. The difference is not mystical; it is architectural.

14.4 Active inference

Active inference and the free-energy principle model organisms as systems that maintain themselves by minimising expected free energy and acting to reduce uncertainty [16]. Mirror Theory is close to this family of views but differs in emphasis. It asks when the reliability of the self-model itself becomes viability-relevant and when governed identity-continuity emerges from that recursive structure. The thermodynamic account developed here is compatible with active-inference style reasoning, but it is not identical to it.

15 What this paper does not claim

For clarity, this paper does not claim any of the following:

- It does not claim that consciousness has been explained.
- It does not claim that artificial agents with memory governance are conscious.
- It does not claim that thermodynamics is modified by observerhood.
- It does not claim that constraint salience is a physical field.
- It does not claim that the speed of light is altered by observerhood.
- It does not claim that resource cost alone is sufficient for observerhood.

The claim is narrower:

If observerhood requires persistent self-modelled identity-continuity, then any physically realised observer must pay the costs of memory preservation, error correction, boundary maintenance, controlled update and reliability governance.

This is a support-condition claim, not a replacement-law claim.

16 Empirical and computational programme

Thermodynamic observerhood suggests several testable directions.

16.1 Artificial agents

In artificial agents, the relevant costs are explicit: compute, memory, latency, verification, tool calls, model querying, storage, bandwidth and repair operations. The Observerhood Labs already show a computational analogue of the thermodynamic salience condition [7–13]. Future work should measure:

- when audit cost exceeds contamination reduction;
- when memory governance improves long-horizon task success;
- when recursive reliability becomes worth its meta-diagnostic cost;
- when commit gates become too conservative;
- how energy or compute budgets shift the observer-threshold phase diagram.

16.2 Biological systems

In biological systems, the relevant costs include metabolism, neural signalling, immune maintenance, repair, sleep, attention and homeostatic regulation. Mirror Theory predicts that self-model

complexity should be constrained by maintenance cost. A biological system should not maintain arbitrarily elaborate self-representations unless those representations improve viability relative to their cost.

16.3 Collective systems

In scientific communities and institutions, memory and error correction are distributed. Archives, peer review, replication, critique and retraction are expensive. They persist because they improve epistemic viability. Mirror Theory predicts that communities with no error correction become contaminated, while communities with excessive audit rigidity become unable to adapt.

17 Toward causal invariance

The next Mirror Physics question concerns causal propagation [5,6]. If persistent observerhood requires ordered update, memory preservation and error correction, then the physical regime must provide reliable causal structure. The invariant speed c in relativity is not merely the speed of light in a naive sense; it is part of the causal architecture of spacetime. Mirror Theory can therefore ask, cautiously, whether finite causal propagation is one of the physical conditions that makes observer-continuity possible.

That future question should not be confused with a claim that Mirror Theory modifies relativity. The present paper only prepares the ground. It shows that observer-continuity is costly and physical. Once that is established, later work may ask how causal order, finite propagation and thermodynamic maintenance jointly constrain the space of possible observers.

18 Conclusion

Mirror Physics III develops the thermodynamic side of observerhood. If an observer is a persistent, self-modelling, reliability-governed process, then it must be physically maintained. Memory must be preserved. Error must be corrected. Boundaries must be maintained. Updates must be governed. Reliability must be audited when necessary. Each of these functions carries cost.

The resulting condition is simple:

$$\Theta_L^\tau(\mu_L) > 0. \quad (15)$$

A maintenance mechanism becomes observer-salient only when the viability it preserves exceeds the physical and computational cost of maintaining it. This makes Mirror Theory explicitly cost-sensitive. It also explains why recursive structure is not always beneficial: recursion costs resources, and resources matter.

The paper therefore gives a disciplined bridge from mathematical observerhood to physical realisation. It does not propose new physics. It identifies what any future physics of observerhood must include: memory substrates, error correction, boundary maintenance, controlled update, thermodynamic throughput and viability-weighted persistence.

A Compact formal summary

A thermodynamic observer-candidate:

$$O_L^\tau = (X_L, B_L, M_L, I_L, R_L, \mu_L, V_L). \quad (16)$$

A memory-stability condition:

$$\mathbb{P}(d(M_L(t), M_L(t + \tau)) < \epsilon) \geq q. \quad (17)$$

A persistence-cost functional:

$$K_L^{\text{persist}} = K_L^{\text{memory}} + K_L^{\text{correction}} + K_L^{\text{boundary}} + K_L^{\text{update}} + K_L^{\text{audit}}. \quad (18)$$

A thermodynamic salience condition:

$$\Theta_L^\tau(\mu_L) = \mathbb{E}[V_L(t + \tau) \mid \mu_L] - \mathbb{E}[V_L(t + \tau) \mid \mu_0] - K_L^{\text{persist}}(t, t + \tau) > 0. \quad (19)$$

A thermodynamic salience correction:

$$\Phi_{L,\text{therm}}^{\Gamma,\tau} = \alpha \mathcal{I}_L^{\Gamma,\tau} + \beta \mathcal{C}_L^{\Gamma,\tau} + \gamma \mathcal{E}_L^{\Gamma,\tau} + \delta \mathcal{R}_L^{\Gamma,\tau} - \lambda K_L^{\text{persist}}. \quad (20)$$

B Glossary

Thermodynamic observerhood

Observerhood considered as physically maintained persistence under cost, noise and constraint.

Persistence cost

The physical and computational cost of preserving memory, correcting error, maintaining boundaries, governing update and auditing reliability.

Functional boundary

A rule, structure or process that distinguishes self-relevant variables from environmental variables for maintenance and action.

Memory governance

The process by which candidate memories are committed, quarantined, audited or rejected relative to identity-continuity and viability.

Audit cost

The cost of checking whether an update, memory or reliability estimate should be trusted.

Thermodynamic salience

The viability advantage of a maintenance mechanism after its persistence cost is subtracted.

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